



Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 416 :: Network Routing and Switching Sessional Project Proposal



Project Title : Design and Implementation of a Scalable, Redundant, and Secure Multi-Site Enterprise Network Architecture (ApexCore)

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Executive Summary

This project proposes the design and simulation of **ApexCore-Net**, a robust, multi-site enterprise network infrastructure engineered to address the performance, scalability, and security challenges of modern business operations. Spanning a Head Office (HO), a Branch Office, and a dedicated Data Center (DC), the proposed architecture implements the classic three-layer hierarchical model (Core, Distribution, Access). Key design implementations include multi-area OSPF routing with route summarization, redundant Layer-2/Layer-3 switching via RSTP, EtherChannel (LACP), and First-Hop Redundancy Protocols (HSRP), alongside structured IPv4 VLSM addressing, NAT/PAT, DHCP, and Access Control Lists (ACLs) for role-based segmentation. The entire network will be implemented and validated using Cisco Packet Tracer.

1 Introduction and Problem Statement

Modern enterprise networks require high availability, predictable throughput, deterministic failover, and solid network-layer security. Traditional flat network topologies suffer from single points of failure, unconstrained broadcast domains, and lack of traffic segmentation.

Deficiencies in the Existing Network:

- **Single Points of Failure (SPOF):** Single gateway routers and unbundled uplinks lead to total site downtime during hardware or link failure.
- **Lack of Segmentation:** Absence of VLANs exposes critical infrastructure to unauthorized lateral access and excessive broadcast overhead.
- **Inefficient Routing & Addressing:** Flat IP assignment causes routing table bloat and wastes IP address space.
- **Security Vulnerabilities:** Unrestricted inter-department traffic and unmanaged external access create significant security risks.

2 Project Objectives

The primary goal is to design, configure, and evaluate an enterprise-grade network simulation meeting the following core objectives:

1. Implement a 3-tier hierarchical network design across Head Office, Branch Office, and Data Center.
2. Develop an optimized IPv4 VLSM addressing plan and basic IPv6 dual-stack allocation.
3. Configure VLANs, 802.1Q trunks, and inter-VLAN routing using Layer-3 switches.
4. Deploy Multi-Area OSPF (Area 0, Area 10, Area 20) with route summarization and authentication.
5. Establish Layer-2 and Layer-3 redundancy using EtherChannel (LACP), Rapid PVST+, and HSRP.
6. Enforce network security using standard/extended ACLs, Port Security, and secure management (SSH).
7. Configure core network services including DHCP (with IP helper), DNS, and NAT/PAT at the Internet edge.
8. Conduct failure-injection testing to verify fault tolerance and recovery behavior.

3 Proposed Network Architecture

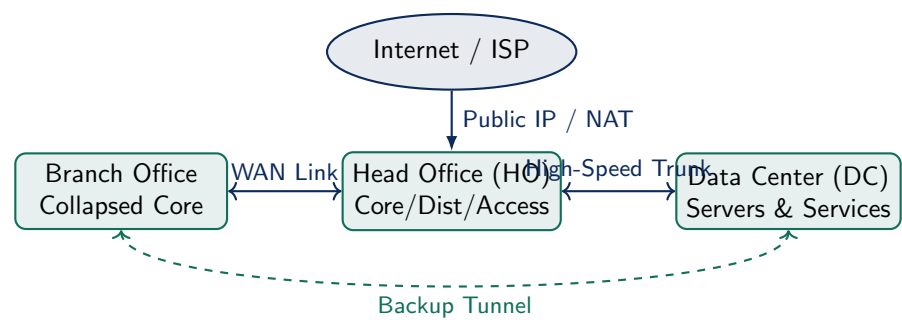


Figure 1: ApexCore-Net High-Level Multi-Site Topology

Site	Architecture	Key Hardware Roles
Head Office (HO)	3-Tier Hierarchy	2 Edge Routers, 2 Multi-layer Switches (Core/Dist), Access Switches
Branch Office	2-Tier Collapsed	1 Integrated Services Router, Access Switches
Data Center (DC)	Redundant Core	1 Gateway Router, 2 Distribution Switches, Application Servers

Table 1: Site Infrastructure Summary

4 Technical Design Specifications

4.1 VLAN Segmentation

To enforce functional boundaries, traffic is partitioned into standard operational VLANs:

VLAN ID	Name	Subnet Block	Access Scope
10	Management	10.10.10.0/24	Network Administrators, SSH access to devices
20	Administration	10.10.20.0/24	HR, Executive, and Office staff
30	Finance	10.10.30.0/24	Accounting and Finance records (Restricted)
40	IT & Engineering	10.10.40.0/24	Technical and Development workstations
50	General Users	10.10.50.0/24	Standard employee workstations
60	Servers (DC)	10.40.60.0/24	Internal Web, Database, and DNS services
99	Guest Wi-Fi	10.10.99.0/24	Isolated Internet-only access

Table 2: VLAN Segmentation and Addressing Framework

4.2 Routing Architecture (OSPF)

The network uses Open Shortest Path First (OSPFv2) as the interior gateway protocol:

- **Area 0 (Backbone):** Connects the Head Office Core and Data Center gateways.
- **Area 10 (HQ LAN):** Covers Head Office campus departments.
- **Area 20 (Branch - Totally Stubby):** Branch subnets receive only a default route (0.0.0.0/0) from the ABR, minimizing memory and CPU utilization.
- **Summarization:** Area Border Routers (ABRs) summarize intra-site subnets into clean /16 advertisements into Area 0.

4.3 Redundancy and High Availability

- **First-Hop Redundancy:** Hot Standby Router Protocol (HSRP) configured on the distribution switch pair to provide transparent default gateway failover.
- **EtherChannel (LACP):** 802.3ad link bundling on inter-switch links to aggregate bandwidth and prevent link-failure convergence penalties.
- **Spanning Tree Protocol:** Rapid PVST+ configured with deterministic Primary and Secondary Root bridges, complemented by BPDU Guard and PortFast on edge ports.

4.4 Security Implementation

- **Access Control Lists (ACLs):** Extended named ACLs deployed on SVIs to enforce least privilege (e.g., denying Guest and User VLANs access to Finance and Server VLANs).
- **Port Security:** Configured on access switchports with maximum MAC thresholds and violation shutdown mode.
- **Device Hardening:** Disabling unused ports, assigning a non-default Native VLAN, enforcing strong password encryption, and enabling SSH (v2) administration.

4.5 Network Services and Edge Connectivity

- **DHCP & Relay:** Centralized DHCP pools for user VLANs with `ip helper-address` configured on Layer-3 switch interfaces.
- **NAT/PAT:** Dynamic Port Address Translation (Overload) on the Edge router allowing internal private pools to access the simulated Internet cloud.

5 Testing and Verification Plan

Test	Scenario	Expected Outcome	Verification Command
1	Core Link Severance	LACP maintains throughput; Rapid PVST+ transitions instantly	<code>show etherchannel summary</code>
2	Active Gateway Failure	HSRP Standby assumes Active state without host packet loss	<code>show standby brief</code>
3	OSPF Link Failure	OSPF reconverges over backup link; routing tables update	<code>show ip route ospf</code>
4	Unauthorized Access	Extended ACL denies forbidden traffic (e.g., Guest → Finance)	Extended ping / Simulation mode
5	Edge NAT Translation	Inside private addresses successfully translate to Edge Public IP	<code>show ip nat translations</code>

Table 3: Failure and Security Verification Matrix

6 Tools and Work Schedule

Simulation Tools: Cisco Packet Tracer (v8.x), draw.io for schematics, and Wireshark / Packet Tracer Packet Inspector for protocol analysis.

7 Conclusion

The proposed ApexCore network architecture provides a complete, scalable, and resilient enterprise networking solution designed to satisfy all academic requirements of the CCE 416 curriculum.

Timeline	Planned Milestones
Weeks 1–3	Requirement specification, topology design, and IP/VLSM address planning.
Weeks 4–6	Layer-2 topology build: VLANs, 802.1Q trunks, EtherChannel, and Rapid PVST+.
Weeks 7–9	Layer-3 routing: Multi-area OSPF deployment, HSRP gateway configuration, DHCP.
Weeks 10–12	Security implementation: Port Security, extended ACL policy matrix, NAT/PAT.
Weeks 13–14	Fault injection testing, convergence measurement, and performance verification.
Weeks 15–16	Final documentation, report submission, and project viva defense.

Table 4: Project Execution Timeline

lum. By integrating multi-area OSPF, robust first-hop redundancy, Layer-2 loop mitigation, and strict ACL-based security, the project will demonstrate both theoretical rigor and practical configuration mastery within Cisco Packet Tracer.

References

1. J. Moy, *OSPF Version 2*, IETF RFC 2328, 1998.
2. Cisco Systems, *Enterprise Campus 3.0 Architecture: Overview and Framework*, Cisco Press.
3. IEEE Standard 802.1D / 802.1w, *Media Access Control (MAC) Bridges and Rapid Spanning Tree Protocol*.
4. Y. Rekhter et al., *Address Allocation for Private Internets*, IETF RFC 1918, 1996.